How-To Library

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Understand Data Observability Anomalies in Data Governance and Catalog

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Abstract

This article explains the various anomalies and their corresponding metrics that data observability in Data Governance and Catalog identifies in your data.

Supported Versions

- Data Governance and Catalog 2024.10.M

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Introduction

Data observability identifies anomalies in your source system data by analyzing sudden deviations from a usual trend.

Data observability classifies the anomalies that it detects into the following categories:

- Percentage-based anomalies
- Count-based anomalies
- Schema-based anomalies

Percentage-based anomalies

Data observability identifies these anomalies when there is a sudden percentage change or percentage deviation in the data.

Percentage-based anomalies contain the following anomaly types:

- Drop from Maximum
- Surge from Minimum
- Percentage Variation
- Deviation

Each percentage-based anomaly type is associated with the following metrics:

- Percentage of Blank Rows
- Percentage of Distinct Rows
- Percentage of Duplicate Rows
- Percentage of Null Rows
- Percentage of Zero Rows

Detection of Drop from Maximum and Surge from Minimum anomalies

To identify these anomalies, data observability requires at least one historic profile run. Perform a minimum of two profile runs on a source system for data observability to detect these anomalies in your source system.

When is the Drop from Maximum anomaly detected?

A drop from 100% to a lower value indicates that there is an anomaly. Data observability identifies these anomalies under the Drop from Maximum anomaly type.

The Drop from Maximum anomaly contains the following levels of severity:

- **Normal Severity**. The value usually remains at 100% but has suddenly dropped between 100% and 95%.
- **High Severity**. The value usually remains at 100% but has suddenly dropped below 95%.

Example 1: Percentage of Distinct Rows

An organization's employee data set includes an `EMP_ID` column with unique or distinct IDs for each employee. Therefore, we expect 100% distinctness in the `EMP_ID` column. If Data Observability detects duplicate employee IDs, the records are no longer 100% distinct. Data Observability then reports that the distinctness value has dropped from the maximum of 100% to a lower value, say 97%. Hence, we see here a Drop from Maximum anomaly.

Example 2: Percentage of Zero Rows

A shopping website sends promotional emails to all its customers each week. In its marketing data set, the company expects the `Customer_Ignored` column to show the 0 value for all its customers. If 100% of its customers have the 0 value in this column, this means that none of the customers were ignored in its promotional emails. If the percentage here drops from 100% to 96%, then it means that 4% of the customers don't have the 0 value, and they were ignored in the promotional offers. This is a Drop from Maximum anomaly.

When is the Surge from Minimum anomaly detected?

A spike from 0% to any higher value indicates that there is an anomaly. Data observability identifies these anomalies under the Surge from Minimum anomaly type.

The Surge from Minimum anomaly contains the following levels of severity:

- **Normal Severity**. The value usually remains at 0% but has suddenly spiked up to 5%.
- **High Severity**. The value usually remains at 0% but has suddenly spiked above 5%.

Example 1: Percentage of Duplicate Rows

A bank's customer data set should have no duplicate records, thus expecting 0% duplicates. If data observability finds duplicate records, it reports the Surge from Minimum anomaly, which indicates an increase from 0% to a higher value.

Example 2: Percentage of Null Rows

An organization's customer data set includes a `NAME` column, where each row much contain a customer's name. We expect the `NAME` column to have 0% null values. If data observability detects any null value, it reports the Surge from Minimum anomaly, which indicates an increase from 0% to a higher value.

Example 3: Percentage of Blank Rows

An organization's customer data set includes an `ADDRESS` column that must contain customer addresses without any blank row. Thus, we expect 0% blank values in the `ADDRESS` column. If data observability detects any blank row, it reports the Surge from Minimum anomaly, which indicates an increase from 0% to a higher value.

Example 4: Percentage of Zero Rows

A retail organization's customer data set includes an `ORDER QUANTITY` column, which contains the quantity of items that a customer ordered, and this quantity should not be 0. We expect 0% of the rows in the `ORDER QUANTITY` column to

contain the value 0. If data observability detects values of 0 in this column, it reports the Surge from Minimum anomaly, which indicates an increase from 0% to a higher value.

Detection of Percentage Variation and Deviation anomalies

To identify these anomalies, data observability requires at least three historic profile runs. Perform a minimum of four profile runs on a source system for data observability to detect these anomalies in your source system.

When is the Percentage Variation anomaly detected?

If the records show the same percentage values for a minimum of three historic profile runs, and the values were not 0% or 100%, then data observability will report any change from these consistent values in any subsequent profile runs. Data observability identifies these anomalies under the Percentage Variation anomaly type.

The Percentage Variation anomaly contains the following levels of severity:

- **Normal Severity**. The value usually remains constant but has suddenly changed between 20% and 50%.
- **High Severity**. The value usually remains constant but has suddenly changed to above 50%.

If the value usually remains constant but has suddenly changed below 20%, then no anomaly is observed.

Note: As part of the sensitivity level, no anomaly will be reported for a change of less than 5%.

Example 1: Percentage of Blank rows

An organization acquires new customers or clients after 5 years and adds their information to an existing data set that was nearly 97% accurate for many years. In this customer data set, some columns, such as `ADDRESS` and `COUNTRY`, might contain blank values because not all information about the new customers is available. This would result in an increase in the number of blank rows. Data observability then reports that there is a variation in the percentage, which has dropped from a maximum of 97% to a lower value, say 70%. Hence, we observe a Percentage Variation anomaly.

When is the Deviation anomaly detected?

If the record's percentage values change with each profile run, the algorithm will calculate an expected range based on the changes observed in each profile run. If a new value falls outside of this expected range, data observability reports it as an anomaly. Data observability identifies these anomalies under the Deviation anomaly type.

The deviation percentage determines the severity of the anomaly, which can be either HIGH or NORMAL. Data observability considers an anomaly when there's a 20% deviation from the expected range.

The Deviation anomaly contains the following levels of severity:

- **Normal Severity**. The value usually falls within a specific range but has deviated by a range between 20% and 50%.
- **High Severity**. The value usually falls within a specific range but has deviated by a range above 50%.

If the value usually falls within a specific range but has deviated by a range below 20%, then no anomaly is observed.

Note: As part of the sensitivity level, no anomaly will be reported for a change of less than 5%.

Example 1: Percentage of null rows

Every day, a retail organization reviews the new customer details in their data set and finds around 5 to 10 rows with null entries, which is around 10% null entries. However, on one particular day, the organization reviewed details of new customers and found around 30 row entries containing null values, which is about 30% null entries. Data observability then reports a deviation from the expected range of 10% null entries every day to 30% null entries on that particular day. Hence, we observe a Deviation anomaly.

Count-based anomalies

Data observability identifies these anomalies based on the count of occurrences where there is a sudden change in the number of events or data points.

Count-based anomalies contain the following anomaly types:

- Drop
- Surge
- Drop in Transition
- Surge in Transition
- Count Variation

Each count-based anomaly type is associated with the following metrics:

- Row Count
- Maximum Length
- Minimum Length
- Average

Detection of count-based anomalies

To identify these anomalies, data observability requires at least three historic profile runs. Perform a minimum of four profile runs on a source system for data observability to detect these anomalies in your source system.

The following metrics associated with count-based anomalies provide column statistics:

- **Row Count**. Indicates the number of rows in a table or data set. The actual metric for Row Count is `numvaluesprofiled`.
- **Maximum Length** or **Minimum Length**. Indicates the maximum or minimum number of characters, including character spaces, for a column or data element.

  For example, if the `Employee_Name` column contains the values "Joe M", "Jane Moss", and "Sacha Heptinstall", the minimum length of the value is five and maximum length of value is 17 characters.
- **Average**. Indicates the average of the values for a numeric column or data element.

The count determines the severity of the anomaly, which can be either HIGH or NORMAL.

The Drop, Surge, Drop in Transition, and Surge in Transition anomalies contains the following levels of severity:

- **Normal Severity**. The value usually remains constant but has suddenly changed between 20% and 50%.
- **High Severity**. The value usually remains constant but has suddenly changed to above 50%.

The severity level for a count variation anomaly is always Normal.

When is the Drop anomaly detected?

A drop from a value that has been increasing to a lower value indicates that there is an anomaly. Data observability identifies these anomalies under the Drop anomaly type.

Example 1: Row Count

In the user data set of a streaming service platform, the number of active user records increases every month at a predictable rate. For example, in January, there were 2.5 million users added to the data set, and in February the number of users increased to 2.52 million users. However, if data observability detects a drop of 70 million users in a particular month, it reports the Drop anomaly, which indicates a drop in the number of active users.

When is the Surge anomaly detected?

A spike from a value that has been decreasing to a higher value indicates that there is an anomaly. Data observability identifies these anomalies under the Surge anomaly type.

Example 1: Row Count

In an organization, the number of new customers was decreasing, but due to a sale held in a particular month the number of new customers increased. If data observability detects an increase in the number of new customers in a month, it reports the Surge anomaly, which indicates a spike in the number of new customers.

When is the Drop in Transition anomaly detected?

A drop from a value that has been decreasing at a constant rate to a significantly lower value indicates that there is an anomaly. Data observability identifies these anomalies under the Drop in Transition anomaly type.

Example 1: Row Count

The `Quantity_Sold` column for a product in a retail organization usually shows a decrease of 10 to 30 units. However, due to the launch of a similar product by a competitor, the number of units sold suddenly decreased by 1,000 units. If data observability detects the sudden decrease in sales, it reports the Drop in Transition anomaly.

When is the Surge in Transition anomaly detected?

A spike from a value that has been increasing at a constant rate to a significantly higher value indicates that there is an anomaly. Data observability identifies these anomalies under the Surge in Transition anomaly type.

Example 1: Row Count

The number of customer records in a data set usually increases by 250 to 560 records. However, during a shopping season, the number of customer records suddenly increased to 3,210 new records. If data observability detects the sudden increase in the number of employee records, it reports the Surge in Transition anomaly.

When is the Count Variation anomaly detected?

A change in a value that is usually constant indicates that there is an anomaly. Data observability identifies these anomalies under the Count Variation anomaly type.

Example 1: Average

An organization tracks the number of days a ticket remains open in the `Days Open` column for customer-reported issues. The organization expects the number of days to be between 11 to 14. If data observability detects an increase in the number of days to 15, it reports the Count Variation anomaly, which indicates a variation in count.

Schema-based anomalies

Data observability identifies these anomalies when there is a sudden deviation in the data structure or format.

Schema-based anomalies contain the Schema Modification anomaly type. Each schema-based anomaly type is associated with the Data Type metric.

Detection of schema-based anomalies

To identify these anomalies, data observability requires at least one historic profile run. Perform a minimum of two profile runs on a source system for data observability to detect these anomalies in your source system.

Data observability considers any change in the schema as potentially significant. It does not consider a pre-defined limit or threshold for schema changes to be insignificant.

The Data Type metric associated with the schema-based anomaly indicates a change in data type for a column or data element.

The severity level for a schema-based anomaly is always Normal.

When is the Schema Modification anomaly detected?

A change in the data type of the values in a column indicates there is an anomaly. Data observability identifies these anomalies under the Schema Modification anomaly type.

Example 1: Data Type

The `Employee_ID` column in an organization stores employee IDs, which are numerical values. The data type of the column is Varchar. Hence, the inferred data type that data profiling generates on the column values is of Integer data type. If a user adds a prefix value, such as “EMP”, to all the IDs in the column, and data profiling generates the data type as String, data observability reports it as a Schema Modification anomaly.

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